Two Proofs of the Matrix-Tree Theorem

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This chapter introduces the Matrix-Tree Theorem and presents two proofs, using induction and the Cauchy-Binet formula, to show how determinants can be used to count spanning trees in a graph.

We denote by $A[i]$ the matrix obtained from A by deleting the i-th row and the i-th column.

1 Kirchhoff's Matrix-Tree Theorem

Theorem (Kirchhoff's Matrix-Tree Theorem).

The number of spanning trees of a graph G is given by

$$\tau(G) = \det(L_G[i]),$$

where i can be any vertex.

Here L_G denotes the Laplacian matrix of G.

2 Fact 1

Let $A \in \mathbb{R}^{n\times n}$, and let E_{ii} be the matrix whose (i,i) entry is 1 and all other entries are 0. Then

$$\det(A + E_{ii}) = \det(A) + \det(A[i]).$$

Consider the permutation expansion definition of the determinant:

$$\det(A = (a_{ij})) = \sum_{\pi} \operatorname{sgn}(\pi) a_{1\pi(1)} a_{2\pi(2)} \cdots a_{n\pi(n)}.$$

This fact is not difficult to see. We replace the (i,i) entry by $a_{ii} + 1$, so we obtain the original determinant plus one extra term.

This extra term corresponds exactly to the sum over all permutations on the set $[n] \setminus \{i\}$, namely the determinant obtained by deleting the i-th row and the i-th column. Hence this term is precisely $\det(A[i])$.

3 First Proof

Now we begin the first proof of Kirchhoff's Matrix-Tree Theorem.

Our first proof proceeds by induction on both the number of vertices and the number of edges of the graph G.

If G is an empty graph with two vertices, then

$$L_G = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Hence $L_G[i] = [0]$, and

$$\det(L_G[i]) = 0,$$

which agrees with the theorem.

Throughout the proof:

- $\tau(G)$ denotes the number of spanning trees of G;
- $G - e$ denotes the graph obtained by deleting the edge e;
- G/e denotes the graph obtained by contracting the edge e, that is, merging its two endpoints into a single vertex.

If i is an isolated vertex, then G has no spanning trees, and the i-th row and the i-th column of L_G are entirely zero. Therefore

$$\det(L_G[i]) = \det(L_{G-i}) = 0.$$

Otherwise, we may assume there exists an edge $e = (i, j)$ adjacent to i. For any spanning tree T, either $e \in T$ or $e \notin T$, so

$$\tau(G) = \tau(G/e) + \tau(G - e).$$

The first term reduces the number of vertices by 1, while the second term reduces the number of edges by 1, so this gives us an induction framework.

First, we establish the relation between L_G and L_{G-e}. Observe that

$$L_G[i] = L_{G-e}[i] + E_{jj}.$$

That is, after deleting the edge e, the only difference between the resulting Laplacian matrix and $L_G[i]$ lies in the degree of vertex j. By Fact 1, we obtain

$$\begin{aligned}
\det(L_G[i]) &= \det(L_{G-e}[i] + E_{jj}) \\
&= \det(L_{G-e}[i]) + \det(L_{G-e}[i, j]) \\
&= \det(L_{G-e}[i]) + \det(L_G[i, j]).
\end{aligned}$$

Here $L_G[i, j]$ denotes the matrix obtained by deleting both the i-th and j-th rows and columns. The last equality holds because once the i, j rows and columns are removed, there is no remaining difference between L_G and L_{G-e}.

Next, we establish the relation between L_G and $L_{G/e}$. Suppose we contract vertex i into vertex j. Then

$$L_{G/e}[j] = L_G[i, j].$$

Hence

$$\begin{aligned}
\det(L_G[i]) &= \det(L_{G-e}[i]) + \det(L_{G/e}[j]) \\
&= \tau(G - e) + \tau(G/e) \\
&= \tau(G).
\end{aligned}$$

The second equality follows from the induction hypothesis, completing the first proof.

4 Fact 2: Cauchy-Binet Formula

Let

$$A \in \mathbb{R}^{n\times m}, \qquad B \in \mathbb{R}^{m\times n},$$

where $m \geq n$. Define:

- A_S: the submatrix of A consisting of the columns indexed by $S \subseteq [m]$;
- B_S: the submatrix of B consisting of the rows indexed by $S \subseteq [m]$.

Let $\binom{[m]}{n}$ denote the collection of all subsets of $[m]$ of size n. Then

$$\det(AB) = \sum_{S \in \binom{[m]}{n}} \det(A_S)\det(B_S).$$

5 Second Proof

Since

$$L_G = \sum_{(i,j)\in E} (e_i - e_j)(e_i - e_j)^T,$$

we may write

$$L_G = BB^T,$$

where $B \in \mathbb{R}^{n\times m}$ is an incidence matrix of G. Each edge (i,j) corresponds to a column vector

$$e_i - e_j.$$

Since $L_G = BB^T$, this once again proves that L_G is positive semidefinite.

If $B[i]$ denotes the matrix obtained by deleting the i-th row of B, then

$$L_G[i] = B[i]B[i]^T.$$

We further let $B_S[i]$ denote the matrix obtained from $B[i]$ by keeping only the columns indexed by $S \subseteq E$.

We also need the following lemma.

Lemma 2.

For $S \subseteq E$ with $|S| = n-1$,

$$|\det(B_S[i])| = \begin{cases} 1, & \text{if } S \text{ is a spanning tree,} \\ 0, & \text{otherwise.} \end{cases}$$

With this lemma, the second proof of the Matrix-Tree Theorem becomes extremely short:

$$\begin{aligned} \det(L_G[i]) &= \det(B[i]B[i]^T) \\ &= \sum_{S \in \binom{E}{n-1}} \det(B_S[i])^2 \\ &= \tau(G). \end{aligned}$$

Here:

- the second equality follows from the Cauchy-Binet formula;
- the third equality follows from Lemma 2.

6 Proof of Lemma 2

Proof.

Assume without loss of generality that the edges in $B_S[i]$ are oriented arbitrarily. That is, we may replace the column corresponding to an edge (i,j) from $e_i - e_j$ to $e_j - e_i$.

This only changes the sign of the determinant, so it does not change the absolute value.

If $S \subseteq E$, $|S| = n-1$, and S is not a spanning tree, then S necessarily contains a cycle. We may orient the edges along this cycle.

If we sum the columns corresponding to this cycle, we obtain the zero vector. Hence the columns of $B_S[i]$ are linearly dependent, and therefore

$$\det(B_S[i]) = 0.$$

Now assume that S is a spanning tree. We prove

$$|\det(B_S[i])| = 1$$

by induction on n.

6.1 Base Case

If $n = 2$, then

$$B_S = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

After deleting one row, we get

$$B_S[i] = [\pm 1],$$

and therefore

$$|\det(B_S[i])| = 1.$$

6.2 Inductive Case

Assume the lemma holds for trees with $n - 1$ vertices.

Let $j \neq i$ be a leaf vertex of the tree, and let (k, j) be the unique edge adjacent to j. Such a leaf vertex exists because every tree with at least two vertices has at least two leaves.

Permute rows and columns so that:

- the edge (k, j) becomes the last column;
- the vertex j becomes the last row.

These permutations may change the sign of the determinant, but they do not affect its absolute value.

Since j is a leaf, the last row has only one nonzero entry. Expanding the determinant along the last row gives

$$|\det(B_S[i])| = \left|\det(B_{S-\{(k,j)\}}[i, j])\right|.$$

By the induction hypothesis,

$$\left|\det(B_{S-\{(k,j)\}}[i, j])\right| = 1.$$

Therefore,

$$|\det(B_S[i])| = 1.$$

This completes the proof of Lemma 2, and hence the second proof of Kirchhoff's Matrix-Tree Theorem.

□